



Holevo Capacity Examples

- Ex-1: Identity (Noiseless) Qubit Channel: $\mathcal{N}(\rho) = \rho$
- Ex-2: Completely Depolarising Qubit Channel: $\mathcal{N}(\rho) = \frac{I}{2}$
- Ex-3: Bit Flip Channel: $\mathcal{N}(\rho) = (1 - p)\rho + pX\rho X$
- Ex-4: Depolarising Qubit Channel: $\mathcal{N}(\rho) = (1 - p)\rho + p\frac{I}{2}$



- Ex-1: Identity (Noiseless)
Qubit Channel: $\mathcal{N}(\rho) = \rho$

Holevo Capacity of Q. Channels.

(i) Noiseless Channel $\boxed{\mathcal{N}(\rho) = \rho}$ $\xrightarrow{\rho} \boxed{\mathcal{N}} \xrightarrow{\rho'}$

Choose orthonormal input states. $\frac{|0\rangle}{\sqrt{2}}, \frac{|1\rangle}{\sqrt{2}}$

$$\rho_0 = |0\rangle\langle 0|, \rho_1 = |1\rangle\langle 1|$$

$$\rho'_0 = \mathcal{N}(\rho_0) = |0\rangle\langle 0| \quad \& \quad \rho'_1 = \mathcal{N}(\rho_1) = |1\rangle\langle 1|$$

Capacity $\rightarrow$ Find Holevo χ of $\left\{ \frac{\rho'_0}{\sqrt{2}}, \frac{\rho'_1}{\sqrt{2}} \right\}$

$$S(\rho'_0) = 0 \quad S(\rho'_1) = 0$$

$$\begin{aligned} \rho &= \frac{1}{2} \rho'_0 + \frac{1}{2} \rho'_1 = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| = \mathbb{I}/2 \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \end{aligned}$$

$$S(\rho) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1 \quad \leftarrow$$

$$\text{Capacity} = S(\rho) - \sum p_a S(\rho'_a) \rightarrow 0$$

$$\rightarrow \boxed{= 1 \text{ bit/qubit ch. use}} \rightarrow \underline{\text{Holevo's Bound.}}$$

$\&$ By using ^upre-shared entanglement + ^u1 use of Noiseless Qubit Channel

$\hookrightarrow$ Superdense Coding. $\equiv$ 2 bit/qubit ch. use Capacity. $\checkmark$



- Ex-2: Completely Depolarising Qubit Channel:

$$\mathcal{N}(\rho) = \frac{I}{2}$$

Ex-2: Completely depolarizing channel $\boxed{\mathcal{N}(\rho) = I/2}$
 Alice sends $\left\{ \underset{\gamma_2}{|0\rangle}, \underset{\gamma_2}{|1\rangle} \right\}$

$$\rho_0 = |0\rangle\langle 0| \quad \rho_1 = |1\rangle\langle 1|$$

$$\left. \begin{array}{l} \rho_0 \longrightarrow \rho_0' = I/2 \\ \rho_1 \longrightarrow \rho_1' = I/2 \end{array} \right\} \text{Holds no information}$$

$$S(\rho_0') = 1, \quad S(\rho_1') = 1.$$

$$\rho = \frac{1}{2} \rho_0' + \frac{1}{2} \rho_1' = \frac{1}{2} \frac{I}{2} + \frac{1}{2} \frac{I}{2} = \frac{I}{2}$$

$$S(\rho) = 1$$

$$\begin{aligned} \chi &= S(\rho) - \sum p_i S(\rho_i) \\ &= 1 - \left(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 \right) = 0 \end{aligned}$$

$C(\mathcal{N}) = 0 \rightarrow$ Channel is destroying all the information.

$\rightarrow$ No distinguishability survives



Ex-3: Bit Flip Channel: $\mathcal{N}(\rho) = (1-p)\rho + pX\rho X$

Ex-3 Capacity of Bit Flip Channel.

$\rightarrow \mathcal{N}(\rho) = (1-p)\rho + pX\rho X \leftarrow$

Alice sending $|0\rangle, |1\rangle$
 $|0\rangle \xrightarrow{X} |0\rangle, |1\rangle \xrightarrow{X} |1\rangle$

$|0\rangle \rightarrow \rho_0 = (1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|$
 $\rightarrow \rho_0 = \begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix}$ ✓

$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$|1\rangle \rightarrow \rho_1 = (1-p)|1\rangle\langle 1| + p|0\rangle\langle 0|$
 $\rho_1 = \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}$

$S(\rho_0) = -p \log p - (1-p) \log (1-p) = H_2(p)$
 $S(\rho_1) = -p \log p - (1-p) \log (1-p) = H_2(p)$

$X|0\rangle = |1\rangle$
 $X|1\rangle = |0\rangle$
 $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 $Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
 $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
 $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$\rho = \frac{1}{2}\rho_0 + \frac{1}{2}\rho_1 = \begin{pmatrix} \frac{1-p+p}{2} & 0 \\ 0 & \frac{p+1-p}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2}I$$

$$S(\rho) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1$$

$$\chi = S(\rho) - \left(\frac{1}{2} H_2(p) + \frac{1}{2} H_2(p) \right) = 1 - H_2(p)$$



Ex-4: Depolarising Qubit Channel: $\mathcal{N}(\rho) = (1 - p)\rho + p\frac{I}{2}$

Ex-4 Depolarizing Channel: $\mathcal{N}(\rho) = (1-p)\rho + p\frac{I}{2}$.

Alice $|0\rangle_{\sqrt{2}}, |1\rangle_{\sqrt{2}} \rightarrow$

$|0\rangle \rightarrow \rho_0 = (1-p)|0\rangle\langle 0| + p\frac{I}{2}$

$$\rho_0 = \begin{pmatrix} 1-p & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} p/2 & 0 \\ 0 & p/2 \end{pmatrix}$$

$$\rho_0 = \begin{pmatrix} 1-p/2 & 0 \\ 0 & p/2 \end{pmatrix} \Rightarrow S(\rho) = H_2(p/2)$$

$|1\rangle \rightarrow \rho_1 = (1-p)|1\rangle\langle 1| + p\frac{I}{2}$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 1-p \end{pmatrix} + \begin{pmatrix} p/2 & 0 \\ 0 & p/2 \end{pmatrix} = \begin{pmatrix} p/2 & 0 \\ 0 & 1-p/2 \end{pmatrix}$$

$$S(\rho_1) = H_2(p/2)$$

$$\rho = \frac{1}{2}\rho_0 + \frac{1}{2}\rho_1 = \begin{pmatrix} \frac{1}{2} - \frac{p}{4} & 0 \\ 0 & \frac{p}{4} \end{pmatrix} + \begin{pmatrix} p/4 & 0 \\ 0 & \frac{1}{2} - \frac{p}{4} \end{pmatrix}$$

$$\rho = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2}I \Rightarrow S(\rho) = 1.$$

$$\chi = 1 - \left(\frac{1}{2} H_2(p/2) + \frac{1}{2} H_2(p/2) \right) = \boxed{1 - H_2(p/2)}$$

Classical Capacity of Phase Flip Channel



Phase Flip Channel

$$\mathcal{N}(\rho) = (1-p)\rho + p Z\rho Z$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$p \in [0, 1].$$

$$\left. \begin{aligned} Z|0\rangle &\rightarrow |0\rangle \\ Z|1\rangle &\rightarrow -|1\rangle \end{aligned} \right\}$$

$$\rho_0 = |0\rangle\langle 0| \rightarrow Z\rho_0 Z = |0\rangle\langle 0|$$

$$\rho_1 = |1\rangle\langle 1| \rightarrow Z\rho_1 Z = |1\rangle\langle 1|$$

$$\mathcal{N}(\rho_0) = (1-p)|0\rangle\langle 0| + p|0\rangle\langle 0| = |0\rangle\langle 0|$$

$$\mathcal{N}(\rho_1) = (1-p)|1\rangle\langle 1| + p|1\rangle\langle 1| = |1\rangle\langle 1|$$

$$S(\rho_0) = S(\rho_1) = 0$$

$$\bar{\rho} = \frac{1}{2}\rho_0 + \frac{1}{2}\rho_1 = \frac{I}{2}$$

$$S(\bar{\rho}) = 1.$$

$$\mathcal{H} = S(\bar{\rho}) - \left(\sum_{i=0}^1 p_i S(\rho_i) \right)$$

$$= 1 - (0) = \boxed{1 \text{ bit/qubit channel}}$$

Same as
Noiseless
Channel



Example: Classical Capacity of Quantum Channel

- **Example:** Pure states qubit system: $\mathcal{L} = \{p_a, |\psi_a\rangle\langle\psi_a|, a = 0,1\}$

- Then Holevo information: $\chi(\mathcal{L}) = S(\rho)$,

- Let $|\psi_0\rangle = \cos \phi |0\rangle + \sin \phi |1\rangle$, $|\psi_1\rangle = \cos \beta |0\rangle + \sin \beta |1\rangle$

- Density Operators:

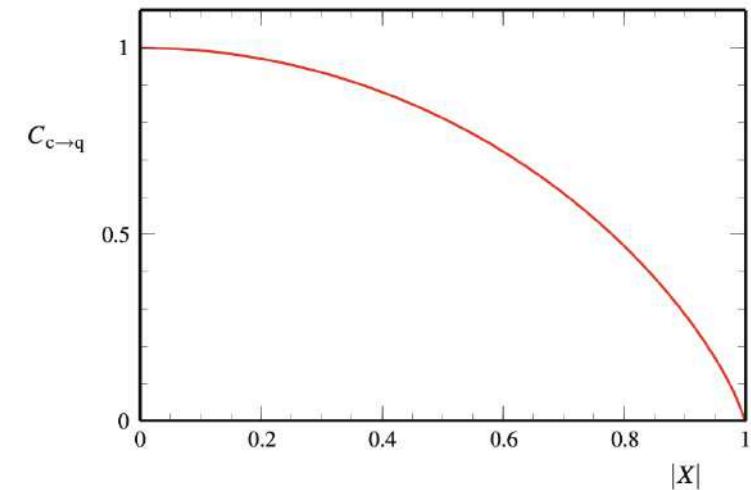
$$\rho_0 = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}, \quad \rho_1 = \begin{bmatrix} \cos^2 \beta & \cos \beta \sin \beta \\ \cos \beta \sin \beta & \sin^2 \beta \end{bmatrix}$$

- Due to symmetry, the optimum is obtained with $p_0 = p_1 = 1/2$.

- Avg. Density operator: $\rho = \begin{bmatrix} \frac{1}{2} [\cos^2 \beta + \cos^2 \phi] & \frac{1}{4} (\sin 2\beta + \sin 2\phi) \\ \frac{1}{4} (\sin 2\beta + \sin 2\phi) & \frac{1}{2} [\sin^2 \beta + \sin^2 \phi] \end{bmatrix}$

- Then $\lambda_{\pm} = \frac{1}{2} \left[1 \pm \sqrt{\cos^2(\beta - \phi)} \right] = \frac{1}{2} [1 \pm |X|]$ where $X = \cos(\beta - \phi)$.

- Thus $C_{c \rightarrow q} = h_2(1/2[1 - |X|])$



Capacity $C_{c \rightarrow q}$ of an input qubit source as a function of the superposition parameter $|X|$ of the input states

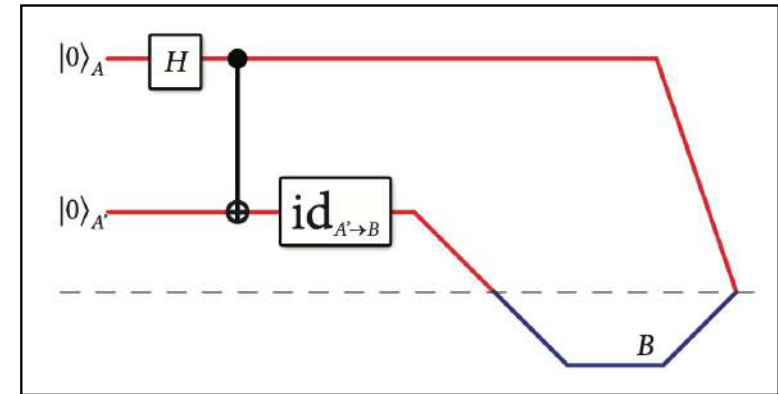


Entanglement Assisted Classical Capacity

Entanglement and Bell State Measurement



- Alice performs local operations (the Hadamard and CNOT) and consumes one use of a noiseless qubit channel to generate one noiseless ebit $|\Phi^+\rangle\langle\Phi^+|_{AB}$ shared with Bob.
- Following states form an orthonormal basis (Bell states):
 $\{ |\Phi^+\rangle_{AB}, X_A |\Phi^+\rangle_{AB}, Z_A |\Phi^+\rangle_{AB}, Z_A X_A |\Phi^+\rangle_{AB} \}$
- So these states are perfectly distinguishable: Using Bell State Measurement (BSM)
- BSM: Convert Bell States $\rightarrow$ Computational Basis
 - BSM performs controlled-NOT from A to B , Hadamard on A , and then measuring A and B in the standard basis.
 - Circuit:
 Bell state $\rightarrow$ CNOT $\rightarrow$ $H \rightarrow \{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$
 - Transformations:
 $|\Phi^+\rangle \rightarrow 00, |\Phi^-\rangle \rightarrow 10, |\Psi^+\rangle \rightarrow 01, |\Psi^-\rangle \rightarrow 11$



Bell States (Orthonormal Basis for 2 Qubits):

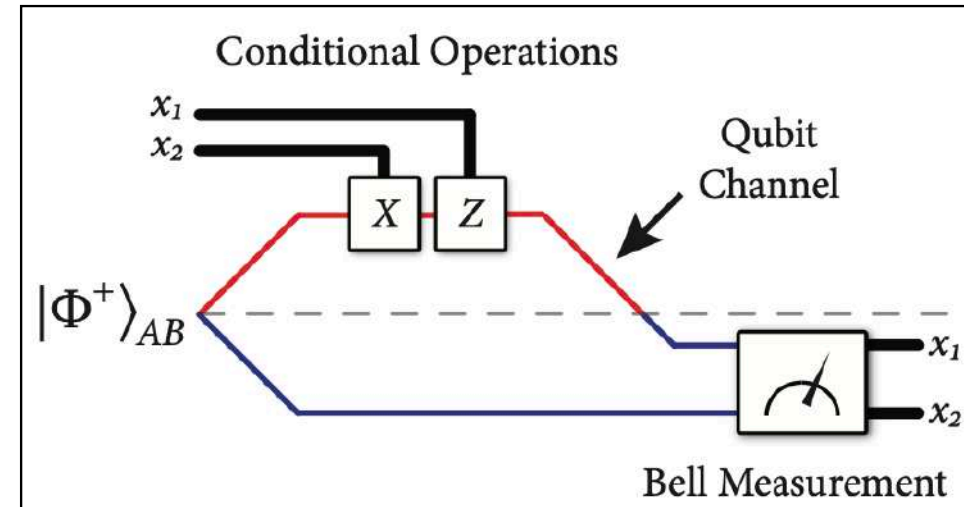
$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$



Superdense Coding

- Holevo Bound: Even though a qubit can exist in infinitely many states, one qubit can reveal at most one classical bit of information (without shared entanglement).
- Superdense coding doubles the classical bitrate by using pre-shared entanglement.
- Alice and Bob share an ebit: $|\Phi^+\rangle_{AB}$
- Alice would like to transmit two classical bits x_1x_2 to Bob
- Alice performs a Pauli rotation conditioned on x_1x_2 :
 $00 \rightarrow I$, $01 \rightarrow X_A$, $10 \rightarrow Z_A$, $11 \rightarrow X_A Z_A$
- Alice sends her share of the ebit over a noiseless qubit channel to Bob.
- Bob then performs a BSM to get x_1x_2
- Following states form an orthonormal basis (Bell states):
 $\{ |\Phi^+\rangle_{AB}, X_A |\Phi^+\rangle_{AB}, Z_A |\Phi^+\rangle_{AB}, Z_A X_A |\Phi^+\rangle_{AB} \}$



Bell States (Orthonormal Basis for 2 Qubits):

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$



Is it Possible to Violate Holevo's Bound?

1 Qubit $\longrightarrow$ max. 1 cbit/ch. use data rate

Q: Why can't encode 2 bits in 1 Qubit

$0 \rightarrow |0\rangle$

$1 \rightarrow |1\rangle$

$\left\{ \begin{array}{ll} 00 \rightarrow |0\rangle & 10 \rightarrow |+\rangle \\ 01 \rightarrow |1\rangle & 11 \rightarrow |- \rangle \end{array} \right\}$

- The 4 quantum states are not mutually orthogonal, hence they can't be perfectly discriminated by Bob.
- Therefore, the two classical bits can't be sent reliably.